complex 2019

QUESTION ONE

(a) Solve the equation $x^2 - 4x + 7 = 0$.

Give your solution in the form $a \pm \sqrt{b}\mathrm{i}$, where a and b are rational numbers.

$$x^2 - 4x = -7$$
$$(x-2)^2 = -3$$
$$x = 2 \pm \sqrt{3}\, i$$

(b) When the polynomial $2x^3 - x^2 - 4x + p$ is divided by $x - 3$, the remainder is 38 .
Find the value of p .

$$f(3) = 38$$
$$2(3)^3 - 3^2 - 4(3) + p = 38$$
$$33 + p = 38$$
$$p = 5$$